

Assignment-5: AJAX, PHP, SQL & Node JS

Time: 1 week

(Visual appeal of your design matters)

Write HTML, JavaScript, PHP and SQL code to enable the client, server and database to communicate. Install any simple SQL software (XAMPP / WAMP / MAMP with MySQL, or equivalent). In each case develop a function and call it.

1. Write a basic Node.js server (using only the built-in http module, no frameworks or Express) that listens on a port and responds with “Hello Node” when the browser requests it.
2. Using your SQL software, create a database and a table named after your own roll number, with at least 4 columns relevant to a topic of your choice (e.g., students, books, expenses). Insert 5 sample records directly through the SQL software (not through code).
3. Write a PHP + AJAX page that fetches all records from your table (Q2) and displays them in an HTML table on the page, without reloading the page.
4. Create a form on the client side containing the following elements: Textbox, Option button (radio), List box, Hidden element. Use AJAX to send the form data to a PHP script that inserts a new record into your table, and display a success or failure message on the same page without reloading.
5. Add a search box above your table (Q3). As the user types a value, use AJAX to send the partially typed input to PHP, which should query the database for matching records and return only the matches. The displayed table should update live as the user types, with no page reload and no submit button.
6. Your table (Q2) should include one numeric column (e.g., marks or amount). Write a PHP function that uses an array to calculate the total and average of that column across all records, and determines a grade or category for each record based on simple conditions you decide (e.g., >80 = A, >60 = B, else C). Display this in a table using AJAX, without reloading the page.

Bonus Assignment: (Carries Extra Marks)

7. Add Edit and Delete buttons/links next to each record in your displayed table (Q3), and use AJAX + PHP to update or delete that record in the database without reloading the page.
8. [Extension of Q5] Make each column header in your displayed table clickable, so that clicking it sorts the table by that column (ascending, then descending on a second click) using AJAX, without reloading the page.